Module 15

1. HTML Basics

Question 1: Define HTML. What is the purpose of HTML in web development?

HTML (HyperText Markup Language) is the standard markup language used to create webpages. It provides the structure of webpages using tags and elements. HTML helps organize content such as text, images, links, forms, tables, and videos on websites.

Question 2: Explain the basic structure of an HTML document. Identify the mandatory tags and their purposes.

The basic structure of an HTML document includes several important tags:

- **DOCTYPE Declaration** – Defines the HTML version.
- **html Tag** – Root element of the webpage.
- **head Tag** – Contains metadata, title, and styles.
- **title Tag** – Displays the webpage title in the browser tab.
- **body Tag** – Contains all visible webpage content.

These tags are necessary for creating a proper webpage structure.

Question 3: What is the difference between block-level elements and inline elements in HTML? Provide examples of each.

Block-Level Elements

- Take full width available.
- Start on a new line.

Examples: div, p, h1, section

Inline Elements

- Take only the required width.
- Do not start on a new line.

Examples: span, a, strong, img

Question 4: Discuss the role of semantic HTML. Why is it important for accessibility and SEO? Provide examples of semantic elements.

Semantic HTML gives meaning to webpage content and improves webpage structure. It is important for accessibility because screen readers can better understand the webpage. It is also important for SEO because search engines can easily identify important webpage sections.

Examples of Semantic Elements

- header
- footer
- main
- article
- section
- aside
- nav

Lab Assignment

Task: Create a simple HTML webpage that includes:

o A header (), footer (), main section (), and aside section ().

o A paragraph with some basic text.

o A list (both ordered and unordered).

o A link that opens in a new tab.

<!DOCTYPE html>

```html
<html>
<head>
  <title>Simple Webpage</title>
</head>
<body>

  <header>
    <h1>Welcome to My Website</h1>
  </header>

  <main>
    <p>
      This is a simple HTML webpage created for practice purposes.
      It contains different HTML semantic elements.
    </p>

    <h2>Ordered List</h2>
    <ol>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ol>
```

```html
        <h2>Unordered List</h2>
        <ul>
            <li>Frontend</li>
            <li>Backend</li>
            <li>Database</li>
        </ul>

        <a href="https://www.google.com" target="_blank">
            Open Google
        </a>
    </main>

    <aside>
        <h3>Aside Section</h3>
        <p>This section contains extra information.</p>
    </aside>

    <footer>
        <p>© 2026 My Website</p>
    </footer>

</body>
</html>
```

2. HTML Forms

Question 1: What are HTML forms used for? Describe the purpose of the input, textarea, select, and button elements.

HTML forms are used to collect user information.

Form Elements

- **input** – Used for single-line data entry.
- **textarea** – Used for multi-line text input.
- **select** – Used to create dropdown menus.
- **button** – Used to submit or reset forms.

Question 2: Explain the difference between the GET and POST methods in form submission. When should each be used?

GET Method

- Sends data through the URL.
- Used for search forms and non-sensitive information.

POST Method

- Sends data securely inside the request body.
- Used for passwords, login forms, and confidential data.

Question 3: What is the purpose of the label element in a form, and how does it improve accessibility?

The label element describes form fields and connects text with form controls. It improves accessibility because screen readers can identify form fields correctly. Users can also click the label to focus on the related input field.

Lab Assignment

• Task: Create a contact form with the following fields:

o Full name (text input)

o Email (email input)

o Phone number (tel input)

o Subject (dropdown menu)

o Message (textarea) o Submit button

```
<!DOCTYPE html>

<html>

<head>

    <title>Contact Form</title>

</head>

<body>


    <h1>Contact Form</h1>


    <form>


        <label for="fullname">Full Name:</label><br>

        <input type="text" id="fullname" name="fullname"

        required minlength="3" maxlength="30"><br><br>
```

```
<label for="email">Email:</label><br>
<input type="email" id="email" name="email"
required><br><br>

<label for="phone">Phone Number:</label><br>
<input type="tel" id="phone" name="phone"
pattern="[0-9]{10}" required><br><br>

<label for="subject">Subject:</label><br>
<select id="subject" name="subject" required>
   <option value="">Select Subject</option>
   <option value="html">HTML</option>
   <option value="css">CSS</option>
   <option value="javascript">JavaScript</option>
</select><br><br>

<label for="message">Message:</label><br>
<textarea id="message" name="message"
required minlength="10" maxlength="200"></textarea><br><br>

<button type="submit">Submit</button>
```

```
    </form>


</body>

</html>
```

Explanation of Form Validation and Labels

Form validation is used to ensure that users enter correct and complete information before submitting a form.

- **required**: Makes a field mandatory. The user cannot submit the form without filling in that field.
- **minlength**: Specifies the minimum number of characters a user must enter.
- **maxlength**: Specifies the maximum number of characters allowed in a field.
- **pattern**: Defines a specific format that the input must follow, such as a phone number or password format.

The **label** element is used to describe form fields.
The for attribute in the label tag is connected to the id of an input field. This improves accessibility and allows users to click the label to focus on the related input field.

Lab Assignment

- **Task: Create a product catalog table that includes the following columns**:
    - Product Name
    - Product Image (use placeholder image URLs)
    - Price
    - Description
    - Availability (in stock, out ofstock)

```
<!DOCTYPE html>

<html>
```

```
<head>
    <title>Product Catalog Table</title>
</head>
<body>

    <h1>Product Catalog</h1>

    <table border="1" cellpadding="10" cellspacing="0">

        <thead>
            <tr>
                <th>Product Name</th>
                <th>Product Image</th>
                <th>Price</th>
                <th>Description</th>
                <th>Availability</th>
            </tr>
        </thead>

        <tbody>

            <tr>
                <td>Laptop</td>
```

```html
    <td>
        <img src="https://via.placeholder.com/100"
        alt="Laptop Image">
    </td>
    <td>₹50,000</td>
    <td>High performance laptop for students and developers.</td>
    <td>In Stock</td>
</tr>

<tr>
    <td>Smartphone</td>
    <td>
        <img src="https://via.placeholder.com/100"
        alt="Smartphone Image">
    </td>
    <td>₹25,000</td>
    <td>Android smartphone with good camera quality.</td>
    <td>Out of Stock</td>
</tr>

<tr>
    <td rowspan="2">Headphones</td>
    <td>
```

```
          <img src="https://via.placeholder.com/100"
          alt="Headphones Image">
        </td>
        <td>₹2,000</td>
        <td colspan="2">Wireless headphones with noise cancellation.</td>
      </tr>

      <tr>
        <td>
          <img src="https://via.placeholder.com/100"
          alt="Headphones Image">
        </td>
        <td>₹2,500</td>
        <td colspan="2">Premium gaming headphones.</td>
      </tr>

    </tbody>

  </table>

</body>
</html>
```

3. HTML Tables

Question 1: Explain the structure of an HTML table and the purpose of each of the following elements: table, tr, th, td, and thead.

Table Elements

- **table** – Creates the table structure.
- **tr** – Creates table rows.
- **th** – Defines table headings.
- **td** – Defines table data cells.
- **thead** – Groups table header content.

Question 2: What is the difference between colspan and rowspan in tables? Provide examples.

Colspan

Colspan is used to merge multiple columns into one cell.

Rowspan

Rowspan is used to merge multiple rows into one cell.

These attributes help create organized and structured table layouts.

Question 3: Why should tables be used sparingly for layout purposes? What is a better alternative?

Tables should not be used for webpage layouts because they reduce flexibility and responsiveness. Modern websites use CSS Flexbox and CSS Grid because they provide better responsive design and easier layout control.

Lab Assignment

- **Task: Create a product catalog table that includes the following columns:**
    - o Product Name
    - o Product Image (use placeholder image URLs)
    - o Price
    - o Description
    - o Availability (in stock, out ofstock)

Additional Requirements:

- o Use thead for the table header.
- o Add a border and some basic styling using inline CSS.
- o Use colspan or rowspan to merge cells where applicable

```
<!DOCTYPE html>

<html>


<head>

   <title>Product Catalog Table</title>

</head>


<body>


   <h1 style="text-align:center; color:darkblue;">

      Product Catalog

   </h1>
```

```
<!-- Table with Border and Inline CSS -->
<table
  border="1"
  cellpadding="10"
  cellspacing="0"
  style="width:100%; border-collapse:collapse; text-align:center;">

  <!-- thead Used for Table Header -->
  <thead style="background-color:lightgray;">

    <tr>
      <th>Product Name</th>
      <th>Product Image</th>
      <th>Price</th>
      <th>Description</th>
      <th>Availability</th>
    </tr>

  </thead>

  <tbody>

    <tr>
```

```
    <td>Laptop</td>

    <td>
        <img
        src="https://via.placeholder.com/100"
        alt="Laptop Image">
    </td>

    <td>₹50,000</td>

    <td>
        High performance laptop for students and developers.
    </td>

    <td style="color:green;">
        In Stock
    </td>

</tr>

<tr>
```

```
    <td>Smartphone</td>

    <td>
        <img
        src="https://via.placeholder.com/100"
        alt="Smartphone Image">
    </td>

    <td>₹25,000</td>

    <td>
        Android smartphone with good camera quality.
    </td>

    <td style="color:red;">
        Out of Stock
    </td>

</tr>

<tr>

    <!-- rowspan Implemented -->
```

```html
    <td rowspan="2">
        Headphones
    </td>

    <td>
        <img
        src="https://via.placeholder.com/100"
        alt="Headphones Image">
    </td>

    <td>₹2,000</td>

    <!-- colspan Implemented -->
    <td colspan="2">
        Wireless headphones with noise cancellation.
    </td>

</tr>

<tr>

    <td>
        <img
```

```
          src="https://via.placeholder.com/100"
          alt="Gaming Headphones Image">
        </td>

        <td>₹2,500</td>

        <td colspan="2">
          Premium gaming headphones.
        </td>

      </tr>

    </tbody>

  </table>

</body>

</html>
```